

REGULATORY SCORECARD

AI Diabetic Retinopathy Screening Tool

Class C · AI-CDS · MD-7 (Central) · likely MD-22 / test-licence (MD-13) path

LIKELY CLASSIFICATION

Class C · AI-CDS

Moderate complexity

CONFIDENCE

Medium confidence



LIKELY PATHWAY

MD-7 (Central) · likely MD-22 / test-licence (MD-13) path

Clinical investigation likely required.

TIMELINE + COST BAND

12–18 months

Likely compliance spend: Rs 17–39L

READINESS



2 / 10

Band: Red (significant work ahead)

RISK LEVEL

High

Risk = potential clinical/regulatory exposure today.

WHAT TRIGGERED THIS CLASSIFICATION

AI-assisted diagnosis

Influences clinical management

Sensitive patient data

Adaptive AI model

TOP GAPS TO CLOSE

No ISO 13485 / IEC 62304 QMS detected

No CDSCO engagement; ACP required for AI/ML SaMD

Limited prospective clinical validation

RECOMMENDED NEXT ACTION

Engage an ISO 13485 + IEC 62304 consultant for a gap assessment and start a QMS build-out plan this quarter.

SECTION 2

Your Likely Regulatory Pathway

Why this likely class applies to your product — and the form sequence reviewers typically expect.

Your product is an AI-driven screening tool that interprets retinal images to flag diabetic retinopathy — a serious, sight-threatening condition where delayed or incorrect detection can meaningfully alter a patient's clinical trajectory. Because the software actively informs a diagnostic decision (rather than simply organising or displaying data), and because the underlying condition is serious, the information it produces likely carries high significance in the clinical workflow. Adaptive AI behaviour adds a further layer of oversight expectation, typically through an Algorithm Change Protocol. Taken together, these factors place the tool in the higher-risk SaMD band under published CDSCO guidance. The likely pathway is MD-7 manufacturing licence via the Central Licensing Authority, preceded by a test licence and clinical investigation.



LIKELY STEP SEQUENCE

01 Test licence (MD-12 / MD-13)

Apply via MD-12 for a test licence (granted as MD-13) so a small number of devices can be made for the clinical investigation.

2–3 MONTHS

TEST LICENCE NOTE

A test licence (granted as MD-13, applied via MD-12) typically precedes the clinical investigation. The investigation itself runs under MD-22 / MD-23 permissions before the device can be cleared for commercial manufacture.

02 MD-22 clinical-investigation permission + EC approval

Design the multi-centre study, get EC clearance, register on CTRI, enrol participants.

9–14 MONTHS

03 MD-7 manufacturing licence (CLA -> MD-9)

File MD-7 with the Central Licensing Authority alongside QMS evidence, the technical file, predicate or substantial-equivalence narrative, and risk file.

6–9 MONTHS

04 ACP / PCCP submission

File the Algorithm Change Protocol alongside the manufacturing licence — modification scope, retraining triggers, validation thresholds, human oversight.

IN PARALLEL

ACP NOTE

Adaptive AI/ML behaviour means an Algorithm Change Protocol (ACP / PCCP) typically accompanies the MD-7 file per the Oct 2025 CDSCO SaMD draft.

SECTION 3

Readiness Gap Analysis

The gaps a reviewer is likely to flag, ordered by priority. Effort and cost bands are typical Indian medtech ranges — your numbers may be lower with in-house resources or higher with a Big-Four-style consultant.

PRI	GAP	WHY IT MATTERS	SUGGESTED NEXT STEP	EFFORT · COST
P1	No ISO 13485 / IEC 62304 QMS detected	For a Class C AI-CDS device, reviewers will likely expect an ISO 13485-aligned QMS with IEC 62304 software lifecycle records — design controls, risk management traceability, CAPA, internal audit, and change control — before a clinical investigation or MD-9 submission is accepted. Without a documented quality system in place, the file is typically held at intake and any clinical investigation timelines may slip until QMS evidence is produced.	Engage an ISO 13485 + IEC 62304 consultant for a gap assessment and start a QMS build-out plan this quarter.	6–9 months · Rs 3–5L
P1	No CDSCO engagement; ACP required for AI/ML SaMD	Based on the Oct 2025 CDSCO SaMD draft, an adaptive AI/ML screening tool like this one is typically expected to file an ACP (or PCCP-style plan) describing the modification scope, retraining triggers, validation thresholds, performance monitoring, and human-in-the-loop oversight before commercial model updates. Without early CDSCO engagement and a drafted ACP, post-market model retraining may trigger fresh approvals each cycle, which can stall product iteration.	Open a CDSCO pre-submission consultation track and draft an Algorithm Change Protocol (ACP) for the retraining lifecycle.	3–5 months · Rs 2–4L
P1	Limited prospective clinical validation	For a Class C diabetic retinopathy AI tool, reviewers will likely require prospective Indian-population clinical evidence — typically a multi-centre study with Ethics Committee approval and CTRI registration — to support sensitivity, specificity, and safety claims across local fundus imaging conditions and patient demographics. This is usually the longest item on the critical path, so delays here may push the overall approval timeline by 9–18 months.	Design an EC-approved prospective validation study on Indian retinal datasets with ophthalmologist-adjudicated ground truth; register on CTRI.	9–14 months · Rs 8–18L

SECTION 4

Timeline + Cost Estimator

Phased roadmap with Indian-context cost bands. Total range and anchor reflect the readiness card's headline estimate.

TOTAL 12–18 months Anchored to typical Class C AI-SaMD timelines: QMS build-out (~4–6 months), clinical validation with EC approval (~6 months) and central CDSCO review (~3–6 months).

Phase 1 — Foundational compliance Engage an ISO 13485 consultant for a gap assessment, document the Intended Use Statement, and lock the classification rationale.	3–6 MONTHS Rs 3–6L
Phase 2 — Technical documentation Build the IEC 62304 software lifecycle file, ISO 14971 risk file, IEC 81001-5-1 cybersecurity controls, and IEC 62366-1 usability evidence.	4–6 MONTHS Rs 3–7L
Phase 3 — Clinical validation Design and run a multi-centre clinical investigation with EC approval and CTRI registration. Typically the longest item on the critical path.	9–14 MONTHS Rs 8–18L
Phase 4 — Submission preparation Assemble the Device Master File, labelling, IFU, predicate or SE narrative, and pre-submission review.	2–3 MONTHS Rs 2–5L
Phase 5 — CDSCO review cycle Submission filing, queries-and-responses cycle, possible site inspection, licence grant.	4–8 MONTHS Rs 1–3L

LIKELY BOTTLENECKS

- Clinical investigation timeline (EC + CTRI + enrolment + follow-up) dominates the schedule.
- Top high-severity gap: No ISO 13485 / IEC 62304 QMS detected.
- ACP filing introduces additional review back-and-forth alongside the main licence.

SECTION 5

Reviewer Insights

What CDSCO reviewers will likely look for in a submission of this shape. Generic regulatory checklists rarely capture reviewer psychology — this is yours.

Strength of clinical validation evidence

Reviewers likely expect prospective Indian-population evidence on fundus images across diverse retinal cameras, skin tones, and DR severity grades, with EC approval and CTRI registration in place before enrolment. Sensitivity and specificity against an ophthalmologist-adjudicated reference standard, pre-specified non-inferiority margins, and the statistical analysis plan typically receive line-by-line scrutiny for a Class C AI-CDS, alongside subgroup performance for referable DR.

Intended Use consistency across documents

Reviewers typically cross-check the Intended Use across the IFU, labelling, marketing site, and classification rationale. For a DR screening tool, drift such as marketing implying diagnostic confirmation while the IFU limits use to triage of referable DR may trigger early queries. Specifying the care setting (primary care vs ophthalmology), operator profile, and that output supports — not replaces — clinician decisions helps keep the file internally consistent.

AI/ML update controls (ACP / PCCP)

Based on the Oct 2025 CDSCO SaMD draft, reviewers likely look for an Algorithm Change Protocol covering retraining triggers (drift on Indian fundus data, new camera onboarding, performance dips by DR grade), locked validation thresholds for sensitivity and specificity, dataset governance, and human oversight gates before any production model update. A PCCP scoping which changes are pre-authorised versus those needing fresh notification typically clarifies the post-market pathway.

Cybersecurity & data handling

Fundus images and linked patient identifiers bring PHI into scope, so reviewers typically ask for a threat model per IEC 81001-5-1, encryption-at-rest and in-transit, role-based authentication for clinic users, audit logging, and a breach SOP aligned to DPDP Act timelines. For cloud inference, CERT-In empanelled hosting evidence, data-residency posture, and a documented vulnerability disclosure process tend to be expected attachments.

Explainability of AI recommendations

Since the tool influences referral decisions for diabetic retinopathy, reviewers typically ask how output reaches the clinician — a per-eye confidence score, heatmaps or lesion overlays highlighting microaneurysms and haemorrhages, image-quality flags, and an explicit ungradable category. Guidance on handling low-confidence or borderline referable cases, plus IFU language reinforcing the ophthalmologist as the responsible decision-maker, helps demonstrate the CDS framing rather than autonomous diagnosis.

SECTION 6

Smart Examples

Annotated good-vs-bad wording. These are snippets, not full forms — copy them as models for your own Intended Use, claim, and risk-justification language.

INTENDED USE **Scoped vs. autonomous Intended Use**

GOOD

The device assists ophthalmologists and trained graders by flagging colour fundus photographs with features suggestive of referable diabetic retinopathy (moderate NPDR or worse) and/or diabetic macular oedema, presented as an advisory output for clinician review in screening settings. It does not perform autonomous diagnosis or referral; the ophthalmologist or trained grader remains the responsible decision-maker.

BAD

The device automatically diagnoses diabetic retinopathy from fundus images and issues the final referral decision for the patient.

WHY THIS IS SAFER

The first phrasing scopes the device as advisory, names the user, and confines the output to referable DR — aligned with the Oct 2025 CDSCO SaMD draft. The second claims autonomous diagnosis and referral, likely pushing classification stricter and inviting oversight questions.

CLAIM WORDING **Performance claim with study basis**

GOOD

In a retrospective evaluation on a held-out test set of [N] fundus images at [centre], graded against a reference standard of [3 retina specialists, adjudicated], the device demonstrated [sensitivity]% sensitivity and [specificity]% specificity for referable diabetic retinopathy at operating threshold [t]. Performance on prospective Indian-population screening data across skin tones and camera models is pending.

BAD

The device matches ophthalmologist-level accuracy in detecting diabetic retinopathy from fundus photographs.

WHY THIS IS SAFER

The first phrasing names study type, sample, reference standard, operating point, and pending Indian data. The second uses an unverifiable comparator ('ophthalmologist-level') and an unbounded target ('detecting DR'). Reviewers typically ask for the underlying study report before clearance.

RISK JUSTIFICATION False-negative risk justification

GOOD

Hazard: missed referable diabetic retinopathy or diabetic macular oedema. Hazardous situation: screener defers re-screening or specialist referral based on a negative device output. Harm: delayed treatment leading to vision loss. Mitigation: output framed as advisory only; workflow requires independent grader review of ungradable/borderline images; operating point tuned to favour sensitivity (typically published sensitivity $\geq$ X% for referable DR); residual risk tracked under PMS and PMCF on Indian screening cohorts.

BAD

The algorithm is highly accurate on fundus images, so missed DR cases are unlikely in screening use.

WHY THIS IS SAFER

The first phrasing follows ISO 14971's hazard-situation-harm chain, names mitigations, and acknowledges residual risk under PMS/PMCF. The second handwaves false-negative risk with no auditable trail — reviewers will likely treat it as a non-answer and request the underlying analysis.